LØSNINGSFORSLAG (SOLUTIONS)

Exam TFE4146 Semiconductor Devices

Wednesday 27th May 2020

Part I

Multiple choice

Q1 - b thermal velocity > drift velocity

Q2 - c Law of mass action $np = n_i^2$
n_i^{Si} @ RT = $1.5 \cdot 10^{10}$ cm^{-3}

Q3 - a The principle of detailed balance

Q4 - c

Q5 - b

Q6 - a $\Phi^{n^+\text{-poly-Si}} \approx \Phi^{n\text{-Si}}$

Q7 - b

Q8 - a The mobile carrier concentration in the inversion channel equals that of the opposite carrier density in the bulk

Q9 - a

Q10 - b

Q11 - a

Q12 - b $S = 2.3 \frac{kT}{q}\left[1 + \frac{C_d + C_{it}}{C_i}\right]$; decreasing the oxide thickness increases C_i and decreasing the substrate doping decreases C_d, both leading to a low value for S

Q13 - a As the base width $W_b \ll L_p^n$ for a p-n-p BJT, we may write

$$\tanh\left(\frac{W_b}{L_p^n}\right) \approx \frac{W_b}{L_p^n} \Rightarrow \gamma = \left[1 + \frac{W_b n_n \mu_n^p}{L_n^p P_p \mu_p^n}\right]^{-1}$$

The 2nd term in this expression for $\gamma \ll 1$, as $W_b \ll L_n^p$, and we have from a 1st order series expansion,

$$\gamma \approx 1 - \frac{W_b n_n \mu_n^p}{L_n^p P_p \mu_p^n}$$

where $L_n^p, \mu_p^n, \mu_n^p, P_p$ are independent of base width and base doping, while $n_n = N_B$

We therefore see that $\gamma_2 < \gamma_1$, since W_b for BJT#2 is twice that of W_b for BJT#1. Moreover, we note from the expression for γ that $\gamma_3 < \gamma_1$, since the base doping N_B in BJT#3 is twice that of BJT#1.

Q14 - C

The smaller the base width, the larger is the (common-emitter) amplification $\beta \approx \frac{\tau_F}{\tau_t}$; thus the larger the inclination of the I_C vs. $-V_{CE}$ characteristics, and the smaller the magnitude of the Early voltage, $|V_A|$.

Q15 - C

$P = FF \cdot V_{OC} \cdot I_{SC}$; both "a" and "c" yield $P = 28\,W$, but a Si PV solar cell cannot have a V_{OC} of 1.4 V since V_{OC} cannot exceed the Si bandgap of ~1.1 eV

Part II

Q1 (1) - electrons

(2) - into

(3) - common-emitter

(4) - $\frac{\tau_n}{\tau_t}$

(5) - electrons

(6) - electrons

(7) - 0

(8) - normal active, cutoff and saturation mode

(9) - both drift and diffusion of electrons

(10) - ohmic losses in the base and reduced effective base width

Q2 This question leaves a lot of room for individual variation in the comments

- The author writes that solar cells are manufactured by joining two separate thin sheets of p- and n-type semiconductor material to form a p-n junction. As we know, p-n junctions are prepared by diffusion or ion implantation of acceptors in n-type substrate – and so that $N_a > N_d$ in the top layer (diffused/ion-implanted region). In PV solar cells, it is usually the other way around, diffusion/implantation of phosphor (a donor) into p-type Si substrate.

- The author claims that p-Si is positively charged, while n-Si is negatively charged, and that electrons jump from the n-side to the p-side when p- and n-type material is joined. As we know, both p- and n-type material is electrically neutral, and the "built-in" potential at the interface results from formation of a bilayer depleted of mobile charge carriers, in which the n-side is positively charged (ionized donors) and the p-side is negatively charged (ionized acceptors). There is otherwise no static electric potential across the interface.

- The section heading "The photoelectric effect" is somewhat misleading, as this effect refers to the emission of electrons from a solid when irradiated with light of frequency $h\nu > \Phi$ (photon energy exceeding the work function of the material).

The term is, however, also used for the development of a voltage V_{oc} over a p-n junction when irradiated by light with a photon energi $h\nu > E_g$ (the semiconductor bandgap). The photons excite electron-hole pairs (EHP), which are separated by the built-in potential in the depletion zone of the p-n junction to set up a static voltage V_{oc} across the device.

- The author writes that the energy in the sunlight (photon energy) is sufficient to "kick" electrons residing in the vicinity of the interface layer across the potential barrier ("built-in" potential?) the so-called bandgap, to form an EHP on each side of the barrier.

Except for the fact that the bandgap is not a potential barrier and that electrons (and holes) do not "jump" across the junction,

but diffuse to the respective edges of the depletion layer and then drift across to the opposide side of the p-n junction in the electric field set up by the depletion layer space charges, we know that the current in a PV solar cell is optically generated reverse current. Thus, the electrons traversing the p-n junction interface pass from $p \rightarrow n$, not from $n \rightarrow p$.

EHPs are formed on both sides of the junction by optical excitation, not an electron on one side bound to a hole on the opposite side of the interface.

Part III

MOSFET - EPROM

If there is no charge deposited on the floating gate electrode, we may consider the device as an ordinary MOSFET with gate oxide thickness $X_{ox1} + X_{ox2}$.

From the textbook formula, we have for the threshold voltage,

$$V_T = V_{FB} - \frac{Q_d}{C_i} + 2\varphi_F$$

where V_{FB} is the flat-band voltage,

$$V_{FB} = \Phi_{ms} - \frac{Q_i}{C_i} = \Phi_{ms} \quad (\text{for } Q_i = 0)$$

Q_d is the depletion charge $[cm^{-2}]$

$\varphi_F \equiv \frac{1}{q}|E_F - E_i|$ equal to the band bending at the onset of inversion in the SiO_2/Si interface

V_T is the threshold voltage at the onset of strong inversion in the SiO_2/Si interface. The depletion charge will then have reached its maximum value (cf. textbook formula),

$$Q_d = -2\sqrt{qN_A\varepsilon_s\varphi_F}$$

As poly-Si and single-crystalline Si can be assumed to have identical work functions, the flatband voltage amounts to,

$$\Phi_{ms} \approx \frac{1}{q}(E_F - E_C) = \frac{1}{q}\left(-\frac{E_g}{2} - q\varphi_F\right)$$

assuming that the Fermi level of n^+ poly-Si coincides with the conduction band edge E_C.

From the textbook, we have for φ_F,

$$\varphi_F = \frac{kT}{q}\ln\left(\frac{N_A}{n_i}\right) = 26\cdot10^{-3}\ln\left(\frac{10^{16}}{1.5\cdot10^{10}}\right)$$

$$= 0.349\ \text{V}$$

The Si bandgap is $E_g = 1.11$ eV. We therefore have

$$\Phi_{ms} = -\frac{1}{2}\cdot 1.11 - 0.349 = -0.904\ \text{V}$$

The depletion charge,

$$Q_d = -2\sqrt{qN_A\varepsilon_r^{Si}\varepsilon_0\varphi_F}$$

$$= -2\sqrt{1.6\cdot10^{-19}\cdot10^{16}\cdot11.8\cdot8.85\cdot10^{-14}\cdot0.349}$$

$$= -4.83\cdot10^{-8}\ \text{C cm}^{-2}$$

The oxide capacitance,

$$C_i = \frac{\varepsilon_i}{d} = \frac{\varepsilon_r^{SiO_2}\varepsilon_0}{X_{ox1}+X_{ox2}}$$

$$= \frac{3.9 \cdot 8.85 \cdot 10^{-14}}{(30+20)\cdot 10^{-7}} = 6.90 \cdot 10^{-8}\ \mathrm{F\,cm^{-2}}$$

And we have for the threshold voltage,

$$V_T = \Phi_{ms} - \frac{Q_d}{C_i} + 2\varphi_F$$

$$= -0.904 + \frac{4.83 \cdot 10^{-8}}{6.90 \cdot 10^{-8}} + 2 \cdot 0.349$$

$$= \underline{\underline{0.494\ \mathrm{V}}}$$

From Gauss' Law

$$\frac{d\mathcal{E}}{dx} = \frac{\rho(x)}{\varepsilon}$$

where $\rho(x)$ is the charge pr. unit area of the gate contact structure, we find for the $\mathcal{E}$-field in the two insulating layers on either side of the floating-gate electrode,

$$\mathcal{E}_2 - \mathcal{E}_1 = \int_1^2 \frac{Q_{fg}\,\delta(x-X_{ox1})}{\varepsilon_i}\,dx = \frac{Q_{fg}}{\varepsilon_i}$$

[Here, $\delta(x)$ is the delta function]

The gate voltage V_G can be expressed as,

$$V_G = V_1 + V_2 + V_S$$

where V_1 and V_2 are the voltages across the two insulating SiO_2 Layers, and V_S is the potential at the SiO_2/Si interface.

Furthermore, we have,

$$V_G = \mathcal{E}_1 X_{ox1} + \mathcal{E}_2 X_{ox2} + V_S$$

$$= \left(\mathcal{E}_2 - \frac{Q_{fg}}{\varepsilon_i}\right) X_{ox1} + \mathcal{E}_2 X_{ox2} + V_S$$

$$= \mathcal{E}_2 (X_{ox1} + X_{ox2}) - \frac{Q_{fg}}{\varepsilon_i} X_{ox1} + V_S$$

This implies a shift in the threshold voltage brought about by the charge Q_{fg} of,

$$\Delta V_T = -\frac{Q_{fg}}{\varepsilon_i} X_{ox1}$$

$$\Rightarrow \quad Q_{fg} = -\frac{\varepsilon_i \Delta V_T}{X_{ox1}} = -\frac{\varepsilon_r^{SiO_2} \cdot \varepsilon_0 \Delta V_T}{X_{ox1}}$$

$$= -\frac{3.90 \cdot 8.85 \cdot 10^{-14} \cdot 8}{30 \cdot 10^{-7}}$$

$$= \underline{\underline{-9.20 \cdot 10^{-7}\ C\,cm^{-2}}}$$

(corresp. to $5.75 \cdot 10^{12}$ electrons/cm^2)

For $V_G = 3V$ without charge on the floating-gate electrode, the transistor operates in saturation, $V_G > V_T$.

$$I_D = \frac{k}{2}(V_G - V_T)^2 = 25\,(3 - 0.494)^2 = \underline{\underline{157\,\mu A}}$$

With a charge $Q_{fg} = 5.75 \cdot 10^{-12}\,q$ (q = electron charge) on the floating-gate electrode, $V_G = 3V < V_T$, and there is no drain current flowing through the transistor, $I_D = 0$. The transistor is in "CUTOFF".